Smart Decentralized Greenhouse – Hardware & Wiring Documentation

Phase 1 – Wiring and Hardware Bring-up

Controller: ESP32-WROOM-32E

Document Scope: Complete electrical specification from the 220-230 V mains source through every ESP32 Pin.

Control Logic: MQTT, and server layers are out of scope for this document

Document Version	1.7
Date	August 18, 2026
Phase	1 out of 9 (hardware bring-up)
Status	Wiring locked – Ready to build

Table of Contents:

1. Introduction

This document presents the complete hardware wiring and electrical architecture of the *Smart Decentralized Greenhous, an academic proof-of-concept prototype* intended for evnetual production deployment.
The system is built around an **ESP32 microcontroller**, powered by a regulated **5 V DC power supply**, and integrates multiple analog and digital sensors, actuators, and control interfaces.
All subsystems have been designed to ensure stable operation, electrical safety, and logical pin allocation for efficient firmware development.

The build or organized as nine dependendency-ordered phases.
Phase 1 – the subject of this document – wires every sensor and actuator to the controller and verifies that each one reads or switches correctly, with no control logic present.

Establishing a clean, verified hardware baseline here removes electrical faults as a variable for every later phase (threshold control, fuzzy arbitration, RBAC, vision, ledger).

The controlling design constraints, all reflected in the pin assignments **that follow, are:**

- **ADC2** is unusable while Wi-Fi is active, so every analog sensor is placed on **ADC1** channel (GPIO 32-39).
- **Strapping Pins** (GPIO 0, 2, 5, 12, 15) are avoided for driven loads, with one deliberate, justified exception (GPIO 5).
- **Input-Only Pins** (GPIO 34, 35, 36, 39) carry only analog inputs, never outputs.
- **All actuators** are binary (on/off), switched trhough an active-LOW relay module, with the 220 V grow-light circuit requiring guaranteed-off behaviour through boot.

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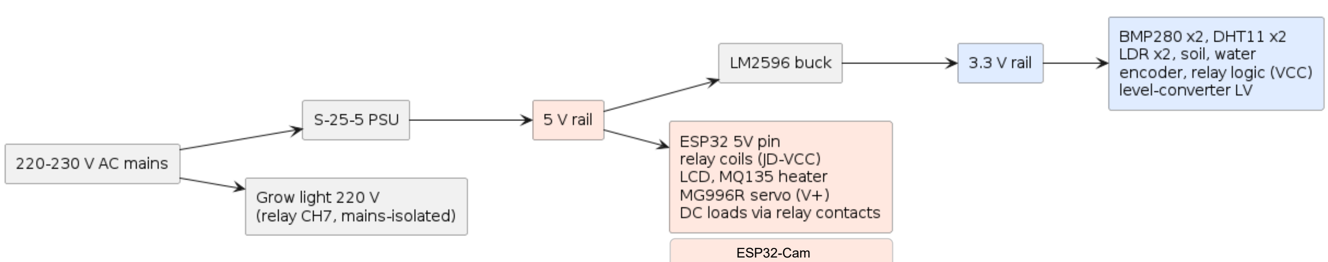
2. System Overview

Phase 1 wires the ESP32 controller to every sensor, the local interface, and every actuator, and verifies each one individually.
No control logic, networking, or server software is part of this phase.

Edge power and signal flow diagram:



The full system splits at MQTT into an edge tier (this controller, must run standalone) and a server tier
This section lists all hardware items used in the Smart CEA prototype, including their electrical type, voltage class, and primary function within the system.

The system also includes the ESP32-Cam, a separate camera node that runs its own firmware and joins wifi independently.

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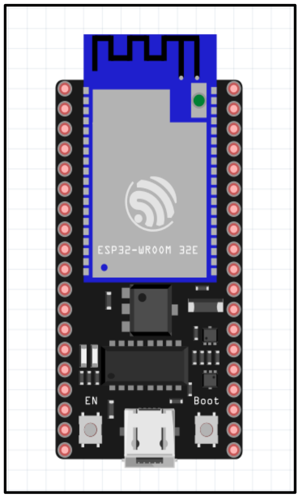
3. Hardware Components & Specifications

3.1. Master Inventory

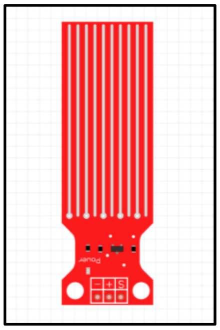
NO.	Component	Quantity	Power Rail	Signal Type	Function	Key Notes
1	ESP32 Wroom-32E	1	5v → onboard 3.3v reg	__	Main Edge-Controller	WiFi/BT ADC2 Unusable with WiFi
2	ESP32 CAM	1	5v independent	__	Camera Node Separate	
3	BMP280 (RobotDyn 7pin)	2	3.3 V	I2C	Inner/Outer Temperature & Pressure	Addr 0x76 / 0x7 No Humidity
4	DHT11 (3-pin module)	2	3.3 V	Digital	Inner/Outer Relative Humidity	±5 % RH (known weak, enough for PoC)
5	MQ135 Gas Sensor	1	5 V	Analog	Air Quality / Gas	Heater Needs 5 V Divider Required
6	LDR Module (KY-018)	2	3.3 V	Analog	Inner/Outer Light Level	Onboard Divider Resistor (No External Needed)
7	Soil Moisture (FC-28 , LM393)	1	3.3 V	Analog	Substrate Moisture	Resistive probe – corrodes under constant bias
8	Water-level Sensor	1	3.3 V	Analog	Reservoir level	Resistive trace; corrodes
9	LCD Display 16x2 I2C Backpack	1	5 V	I2C	Local status display	Addr 0x27/0x3F Via level converter
10	Bidirectional logic level converter	1	3.3 V + 5 V	__	I2C Voltage bridge for LCD	MOSFET (BSS138) Type
11	Rotary Encoder (EC11/KY-040)	1	3.3 V	Digital	Local UI Input	Needs pull-ups (internal used)
12	8-Channel Relay Module	1	5 V Coils 3.3 V Logic	Digital (Active-LOW)	Actuator Switching	SONGLE SRD-05VDC JD-VCC jumper
13	MG996R Servo	1	5 V	PWM (50Hz)	Positional Actuator	High-torque; stall ~2.5 A only PWM load
14	DC Fans (5V 0.08A)	3	5 V (Switched)	__ (Load)	Cooling / Humidity Exhuast	Staged Ventilation Fans (0, 1, 2, 3)
15	DC Submersible Pump	1	5 V (Switched)	__ (Load)	Irrigation	
16	Ultrasonic Humidifer	1	5 V (Switched)	__ (Load)	Humidity Addition	Driver inrush measure draw
17	Led String	1	5 V (Switched)	__ (Load)	Inner Lighting	Normal Lights
18	Led Grow Light	1	220 V (Switched)	__ (Load)	Photoperiod Lighting	Non-Dimmable Mains-isolated

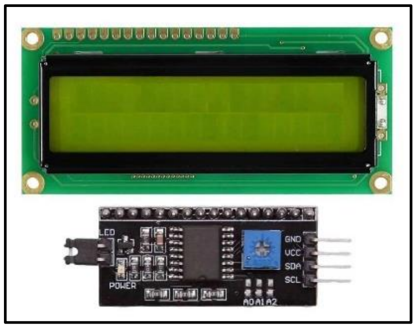
19	S-25-5 PSU	1	220 V in / 5 V out	__	Primary 5 V Supply	25 W, 5 A
20	LM2596	1	5 V in / 3.3 V out	__	3.3 V rail generator	Adjustable; Set to 3.3 V
21	Resistors	9	__	__	Passives	To be detailed later

Components

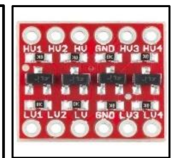


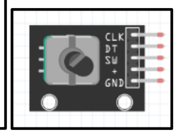
(1) ESP32-Wroom-32E


(8) Water Sensor

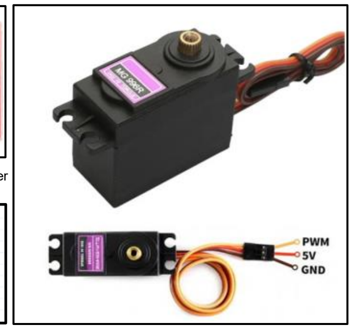


(9) LCD 16x2 & I2C

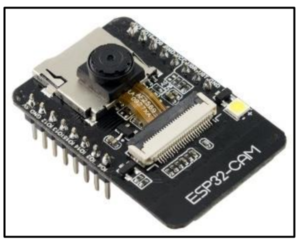

(10) Logic Level Converter



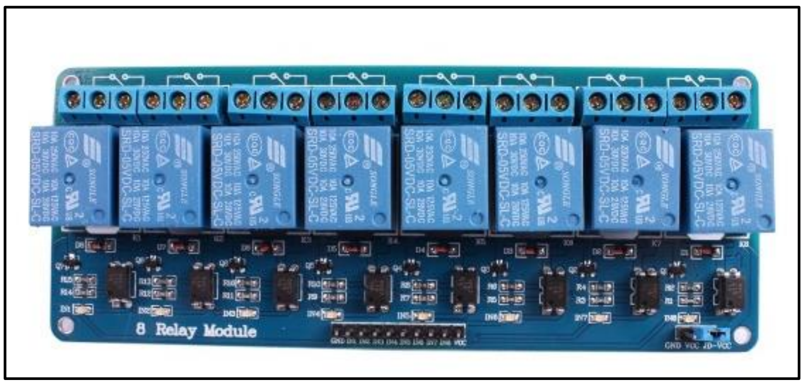
(11) Rotary Encoder



(13) MG996R Servo Motor

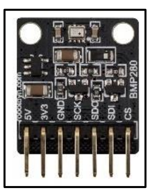


(2) ESP32 CAM

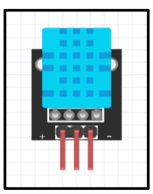


(12) 8-Channel Relay Module


(14) DC Brushless Fans

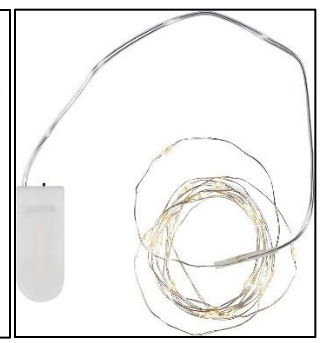


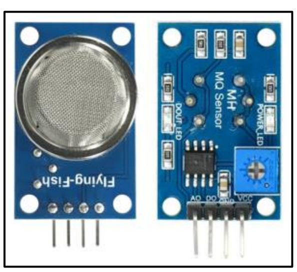
(3) BMP280


(4) DHT11

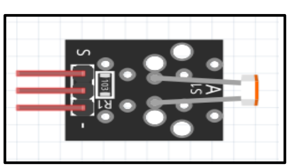

(15) DC Submersible Pump

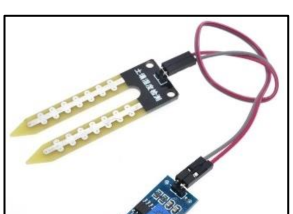

(16) Ultrasonic Humidifier


(17) Micro Led String

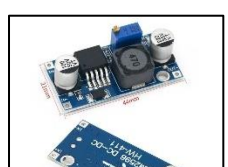


(5) MQ135


(6) LDR



(19) S-25-5 Power Supply Unit



(7) Soil Moisture (18) Led Grow Light (20) LM2596 HW-411 (21) Resistor

3.2. Extended Component Notes

— **ESP32-WROOM-32E**

Dual-Core Xtensa LX6 @ 240MHz, 2.4 GHz WiFi + BT, 3.3V Logic

Powered on the bench via USB (5V → onboard AMS1117 → 3.3V)

For standalone operation the 5V rail feeds the 5V pin.

Two ADC Blocks exist, but ADC2 shared hardware with the WiFi radio and returns gargbade once WiFi is up – hence the ADC1-only rule for all analog sensors.

— **ESP32-Cam**

A separate camera-equipped micro-controller that runs its own firmware and joins WIFI independently.

It has <u>no data connection to the ESP32-WROOM-32E</u>**,** no shared GPIO or signal lines, but it is powered from the shared S-25-5 V Rail. The Board regulates to 3.3 V internally.

— **BMP280 (7-pin RobotDyn Variant)**

Exposes (5V – 3V3 – GND – SCK – SDO – SDI – CS) and supports both I2C and SPI.

For I2C it requires two deliberate pin states:

CS tied High; selects I2C Mode

SDO Selects the address (GND → 0x76 / 3.3V → 0x77)

Exactly how the two sensors are diffrentiated on a shared bus.

Measures pressure (300 – 1100 hPa) and Temperature (-40 – +85 °C) deliberately provides no humidity.

— **DHT11**

Humidity (20 – 90 % RH) at ±5 %; integer resolution.

The 3-Pin Module carries an onboard pull-up, so no external resistor is needed.

Powered at 3.3V so its open-drain data line idles at an ESP32-Safe Level.

— **MQ135**

Tin-Dioxide gas sensor; its internal heater requires 5V, so it cannot run at 3.3V.

The analog output (AO) can swing to nearly the 5V rail, so it passes through a divider before reaching the ADC.

Requires 24-48 h burn-in for stable readings.

The 5V digital output (DO) is left un-connected.

— **LDR / Soil / Water Modules**

All three are breakout modules with onboard signal conditioning that output a 0-toVCC Analog Voltage.

Powered at 3.3V, their outputs already land within the ADC range, so none of them needs an external divider.

The soil (Resistive FC-28) & Water-level (Resistive-Trace) probes corrode under continuous DC bias and should not be left energised for long periods.

— **LCD 16x2 & Level Converter**

The HD44780 + PCF8574 backpack requires 5V for readable contrast; at 5V its onboard pull-ups drive SDA/SCL to 5V, which the **ESP32 is not 5V tolerant for.** The bi-directional level converter bridges only the LCD's two I2C lines between the 3.3V and 5V domains.

The BMP280s are native 3.3V and connect directly to the bus; they do not pass through converter.

— **8-Channel Relay Module**

Opto-isolated, active-LOW (GPIO LOW ~ Relay State On).

SONGLE SRD-05VDC-SL-C relays, contacts rated ~ 10A @ 250VAC / 30VDC.

The board carries a removable VCC → JD-VCC Jumpe: removing it lets the coils run from 5V (JD-VCC) while the opto-input logic side runs from 3.3V (VCC), so a 3.3V GPIO HIGH fully releases the relay.

— **MG996R Servo**

High-torque metal-gear positional servo, 4.8-7.2V, controlled by a 50Hz PWM Signal

(~500-2500 µs pulse → 0–180°).

Idle draw is small, but stall current can reach ~ 2.5A, so it is treated as a transient high-current load on the 5V rail and given a bulk capacitor.

The 3.3V PWM signal from the ESP32 is normally sufficient (logic threshold ~2.5V).

It is only PWM-Driven actuator; all others are binary relay loads.

— **Power Supplies**

The S-25-5 Power Supply Unit (PSU) converts mains to 5V at up to 5A.

The LM2596 HW-411 buck steps 5V down to a regualted 3.3V (~1.7V headroom, adequate).

Warning: Set and verify the LM2596 output with a multimeter before connecting any 3.3V device!

4. Electrical Color Coding

To maintain consistency and safety across all modules, makes the harness self-documenting and reduces misiwiring, the following color code standard is used throughout the circuit:

Color	Net	Voltage Level
	+5V	5V DC
	+3.3V	3.3V DC
	GND	-
	SDA	3.3V logic
	SCL	3.3V logic
	Analog Signal	0-3.3V

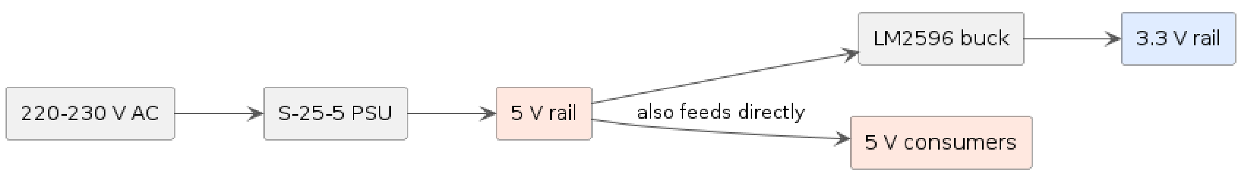
	Digital Signal	3.3V logic
	PWM – Control Output	3.3V logic
	Configuration	-

Notes:

- AC main wiring is intentionally excluded from this color convention.
  The **Live** / **Neutral** / **Earth** conductors are distinguished by their heavier cable gauge and physical separation, not by color, so no DC/mains color collision applies.
  Mains isolations requirements are covered further.
- Green has one exception: The MQ135 signal upstream of its divider carries up to 5V; it only becomes 0–3 V after the divider.
- Purple is the control pulse only: The MG996R Servo power is a separate red 5V wire; the purple wire carries just the 3.3V PWM signal.

5. Power Supply Architecture

5.1. Distribution Chain



Stage	Input	Output	Capacity
S-25-5 PSU	220 – 230 V AC	+5 V / GND	5 A (25 Watts)
LM2596 Buck	5V	+3.30 V / GND	~ 2–3 A (Set by module)

5.2. Rail Allocation

- **220V mains:**
  - The S-25-5 input
  - The grow light (through CH7 relay contacts, isolated)

- **5V rail** powers:
  - ESP32-Wroom-32E 5 V Pin (Standalone only)
  - ESP32-Cam
  - Relay JD-VCC (coils)
  - LCD VCC

- MQ135 VCC
- MG996R Servo
- All Switched DC Loads (3 Fans, Pump, Humidifier, LED String) through the relay contacts
- LM2596 Buck

▪ **3.3V rail** powers:
- Both bmp280
- Both DHT11
- Both LDR
- Soil module
- Water Sensor
- Rotary Encoder
- Relay Logic VCC

5.3. Current Budget (Estimates – Verified by measurement)

Rail	Consumer	Approximately Current
5V	ESP32	~250mA – Peak ~500mA
5V	ESP32-Cam	~0.2-0.3 A
5V	Relay Coil (6 x ~80mA)	~560mA
5V	MQ135 Heater	~150mA
5V	LCD	~25mA
5V	LM2596 input (feeds 3.3V Rail)	~200mA
5V	MG996R Servo	0.7A
5V	3 DC Fans	~450Ma
5V	DC Pump	~300mA
5V	Ultrasonic Humidifier	~300mA
5V	LED String	~200mA
Worst Case Total		**~5A**
3.3V	All Sensors + Encoder	~50mA
3.3V	Relay opto inputs (7 x ~5mA)	~35mA
Worst Case Total		**~85-100mA**

5.4. Powering the ESP32 – bench vs standalone

- **Bench / bring-up** – power the ESP32 from the USB, Do not also feed the 5V rail to the 5V pin simultaneously (however if an internal on-board diode exists, it has no problem, it has been physically tested, but not adviced to do so in the regular cases, however, EXPRESSIF Says it does)
- **Standalone** – connecting the 5V rail to the ESP32 5V pin
- **Note**
  Not to feed-back the 3V3 pin at all, it fights the onboard-regulator.

6. Grounding Topology

A single common ground ties together:

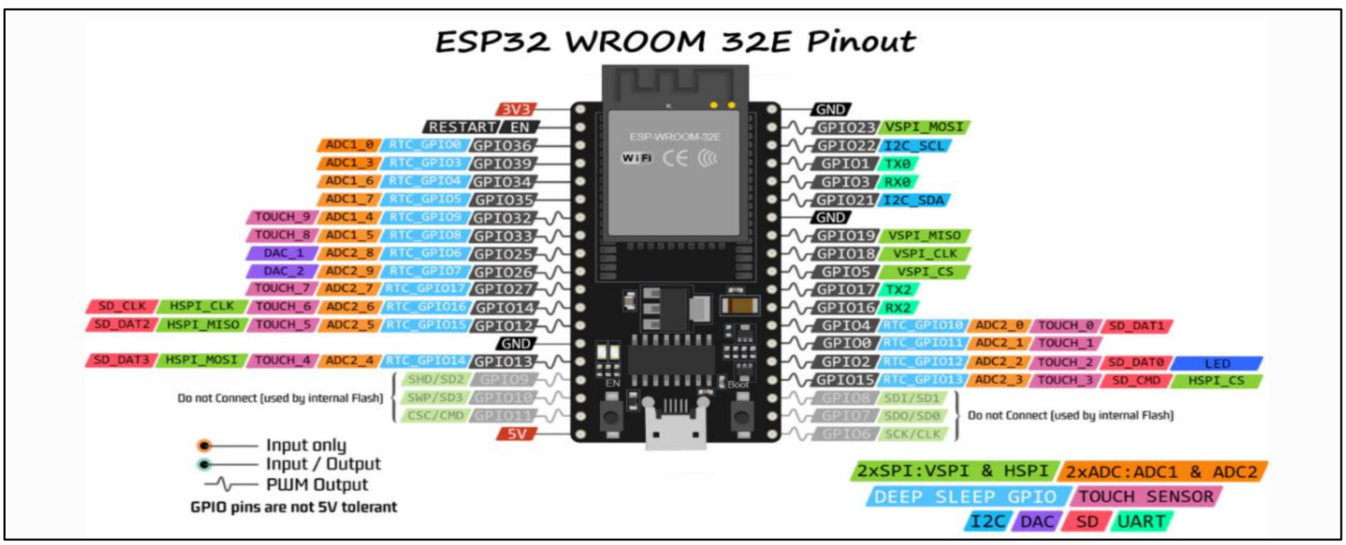
S-25-5 GND , LM2596 I/O GND , ESP32 GND , Relay Logic GND , Level-Converter GNDs , Servo GND , and every module GND.

Route grounds back toward a single node (star topology) rather than daisy-chaining through high current paths; so actuator and servo swithching currents do not corrupt sensor reference voltages.

The one exception: the relay's output contacts for the grow light carry 220V and are galvanically separate dry contacts. That mains side never connects to logic ground.

7. ESP32 Pin Architecture

7.1 Design rules applied



Pin Type	Pins	Description
Power Supply	3V3 , GND	Powers the module (up to 500mA peak during WiFi)
Enable / Reset	EN (Chip_PU)	High → Enables Chip , Low → Hardware Resets
Input Only	GPIO 34,35,36,39	Input only pins, also ADC1 Channels, cannot drive outputs and have no software selectable pull-ups

General Purpose I/O	GPIO 4,13,14,16,17,18,19,21,22,23,25,26,27,32,33	Bidirectional digital I/O support PWM,ADC,DAC,Touch,SPI, I2C, UART
I²C Default	GPIO21 (SDA) , GPIO22 (SCL)	Shared Bus for LCD and BMP280 Sensors
Boot Sensitive Pins	GPIO: 0,2,5,12,15	Avoided – Unused Pins
Restricted Pins	GPIO: 6,7,8,9,10,11	Flash Memory Bus – NOT USED!
UART Pins	GPIO1 (TX0) , GPIO3 (RX0)	Programming & Debugging Purposes

— **General Hardware Rules**

- **Logic level –** All GPIO are 3.3V; applyying 5B damages the chip!
- **ADC2 vs Wi-Fi –** ADC2 pins (GPIO 0,2,4,12,13,14,15,25,26,27) cannot be read while the Wi-Fi is active. Using ADC1 (GPIO 32-39) for continuous analog reads.
- **Boot-Modepins –** at reset GPIO 0 & 15 must be high, GPIO 12 must be low, GPIO 2 must be low/floating, Don't let anything wired to them fight those levels, or the board will not boot.

7.2 Notes – This project's specific use

- **ADC1 fully used for analog**
  GPIO 32,34,35,36,39 carry the analog sensors (MQ135 / Soil / Water / LDR_{x2})

- **GPIO33 borrowed as a digital input**
  although it is an ADC1 pin, here it is the rotary encoder switch (SW) – a digital input, not analog.

- **Digital-input Group**
  encoder CLK/DT/SW on GPIO 13/14/33 , $DHT11_{x2}$ on GPIO 26/27
  all using internal pull-ups.

- **Output Group**
  relay channels IN1→IN7 on GPIO 4/5/16/17/18/19/23 , MG996R servo PWM on GPIO25.

- **Strapping-pin Exception**
  GPIO5, it is the one strapping pin we drive a relay output. It is safe because it idles high (relay off) and its boot glitch only affects a fan/light briefly.
  GPIO 0/2/12/15 are left unused. This is why the general "Avoid Strapping Pins" rule has exactly deliberate exception in our design.

- **Relay boot-safety ties to the boot-mode rule**
  every relay line gets a 10 kΩ pull-up to 3.3 V and it is driven high first in firmware.

- **I²C bus is shared**
  GPIO 21/22 carry both BMP280s (3.3V Side) and the LCD (5V side, via level converter).

7.3 Sensor wiring configuration

The following subsections cover everything wired to the main ESP32-Wroom-32E.

Instruction: in every table connect the module's power pin to the indicated rail, its ground pin to the common GND, and its signal pin the listed ESP32 GPIO.

7.3.I Analog Sensors

All analog sensors sit on ADC1.

The four input-only pins (34/35/36/39) carry the passive modules; GPIO 32 carries the MQ135. Only the MQ135 needs a divider.

Sensor	Power	GND	Signal → ESP32	Notes
LDR Inner (KY-018)	3.3V	GND	S → GPIO36	Onboard divider
LDR Outer (KY-018)	3.3V		S → GPIO39	Onboard divider
Soil Moisture (FC-28)	3.3V		AO → GPIO34	DO un-connected
Water Level	3.3V		S → GPIO35	--
MQ135	5V		AO → Divider → GPIO32	DO un-connected

7.3.II Digital Sensors

Sensor	Power	GND	DATA → ESP32	Notes
DHT11 #1 (INNER)	3.3V	GND	GPIO26	--
DHT11 #2 (OUTER)	3.3V		GPIO27	--

7.3.III I²C Bus

One bus (GPIO 21 = SDA , GPIO 22 = SCL) carries both BMP280s & the LCD, split into two voltage domains by the level converter.

Domain	Devices	Connection
LV 3.3V	ESP32 + BMP280x2	— Direct to GPIO 21/22 — Converter LV Channels

HV 5V	LCD Only	Converter HV Channels

— BMP280x2 Wiring (the two differ only on SDO)

Module pin	Inner (0x76)	Outer (0x77)
3V3	3.3V	3.3V
5V	--	--
GND	GND	GND
SCK	GPIO 22 (SCL)	GPIO (SCL)
SDI	GPIO 21 (SDA)	GPIO 21 (SDA)
SDO	GND (sets 0x76)	3.3V (sets 0x77)
CS	3.3V (I^2C mode)	3.3V (I^2C mode)

— Level Converter & LCD Wiring

Converter pin	Connects to
LV	3.3V
HV	5V
GND (Both HV & LV)	Common GND
LV1	LV1 → GPIO21 (Shares with BMP280s the SDA)
LV2	LV2 → GPIO22 (Shares with BMP280s the SCL)
HV1	HV1 → LCD SDA
HV2	HV2 → LCD SCL
LCD (VCC/GND)	5V/GND

Notes:

Pull-ups: the BMP280 borads carry onboard I^2C pull-ups

Expected addresses on the bus: 0x76 , 0x77 , and the LCD at 0x27 (or 0x3F)

7.3.IV Actuators & Outputs

— Relay Module

IN	ESP32	Load	Contact Wiring (COM / NO)		
			COM	NO	GND
IN1	GPIO19	Pump	5V	Pump (+)	Pump (-)
IN2	GPIO18	S-Fan	5V	Fan (+)	Fan (-)
IN3	GPIO5	Internal Fan	5V	Fan (+)	Fan (-)

IN4	GPIO17	N-Fan	5V	Fan (+)	Fan (-)
IN5	GPIO16	Humidifier	5V	Humi (+)	Humi (-)
IN6	GPIO4	Led String	5V	Led (+)	Led (-)
IN7	GPIO23	Grow Light	Mains (L 220V)	Light (L)	Mains (N) → Light (N)

Note: GPIO5 is a strapping pin used deliberately for the Internal Fan.

— **Servo Motor**

The MG996R is the only PWM-Controlled actuator in the build.

It is driven by a 50Hz signal on a single pin; power and ground come from the 5V rail and common ground.

Servo Wire	Connects to	Note
Brown (GND)	Common GND	--
Red (V+)	5V Rail	4.8-7.2 V (High Transient/Stall Current)
Signal	GPIO25	50Hz PWM (LEDC/ESP32 Servo)

Configuration notes:

- Place a bulk capacitor (470-1000µF) across the servo V+/GND, physically close to the servo, to absorb inrush and protect the shared 5V rail.
- Neveer power the servo from the ESP32 3V3 pin.
- The 3.3V PWM signal from GPIO25 is normally sufficient servo logic threshold ~2.5V).
  if positioning is jittery, insert a 3.3V → 5V level shift on the signal line.
- GPIO25 is full-GPIO and LEDC-PWM capable, it is the only output pin used for PWM.
- On boot the firmware deives the servo to a known neutral angle (90°).
  Set this to you mechanism's safe rest position.

7.3.V Human Interface (Rotary Encoder & LCD)

Rotary encoder uses the ESP32 internal pull-ups (no external resistors).

Must be on full-GPIO pins (no the input-only 34,35,36,39)

Encoder pin	Power/GND	ESP32
VCC+	3.3V	--
GND	GND	--
CLK	--	GPIO13
DT	--	GPIO14
SW	--	GPIO33

Note:

for the LCD wiring see section – 7.33.III

7.3.VI **Reserved Pins**

Pin	Description	Status
GPIO1 (TX0) , GPIO3 (RX0)	USB-UART Interface	Reserved for Programming & Debugging
GPIO6 → GPIO11	SPI Flash Interface	Unusable / Reserved
GPIO 0/2/12/15	Boot strap / Flash Voltage	Avoided
GPIO5	Strapping pin / safe for a fan load	Justified
3V3	Onboard regulator output	FREE

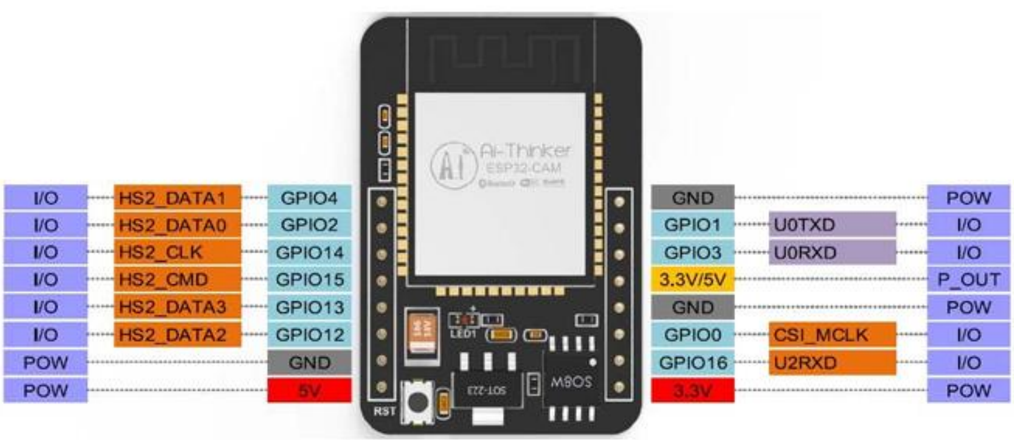
Note:

All other usable GPIOs are assigned, no free full GPIO pins remain after the servo on GPIO25.

8. ESP32-Cam

The ESP32-Cam is the system's second controller, a camera node for the vision layer.

In this phase it is present and powered on the board but exchanges **no data** with the ESP32-Wroom-32E, the two share only the power rails and will coordinate over the network in a later phase.



Schematic Diagram

8.1 Overview

- AI-Thinker ESP32-Cam (Could be integrated) an ESP32 Module with an OV2640 camera and a microSD Slot, and an onboard 3.3V regulator (accepts 5V supply).
- It runs its own firmware and joins Wi-Fi independently.
- It has no USB Port – it is flashed through an external USB-to-TTL (FTDI) adapter.
- None of its pins connect to the main controller.

8.2 Power

Cam pin	Connects to	Note
5V	5V rail (S-25-5)	Preferred input, onboard regulator produces 3.3V
GND	Common GND	Shared reference

The ESP32-Cam is brownout sensitive, give it a solid 5V feed and a decoupling capacitor (100μF at least, 470μF preferred) close to its 5V pin.
Transients on the shared rail (servo stall, pump inrush) can otherwise reset it.
Never power it from the Wroom-32E's 3V3 pin.

8.3 Programming (Flashing)

Cam pin	USB-TTL (FTDI)
5V	5V
GND	GND
U0T (GPIO1)	RX
U0R (GPIO3)	TX
GPIO0	GND – Only while flashing

Note:
Tie GPIO0 to GND to enter download mode, upload the firmware, then remove that link and reset for normal operation.

8.4 Data link to main controller

None in this phase.
The ESP32-Cam & the ESP32-Wroom-32E are not wired together and no shared GPIO, I^2C, or UART, and no pin on the main controller is allocated to the camera.
Coordination between the two runs over the network and belongs to a later phase.

9. Boot Safety & Critical Design Notes

Relay boot safety

Every ESP32 GPIO is high impedance from reset until firmware runs.

On an active-LOW board a floating input can drift LOW = relay ON.

Three layers prevent this, mattering most for the 220V grow light.

— 10 kΩ pull-up from each IN line to the 3.3V logic / VCC hold IN High (OFF) during the float windows.

— Firmware drives every relay pin OUTPUT & High as the first action in setup(), before Serial, I2C, or WiFi.

— Relays are kept off strapping pins (Except for GPIO5, justified below)

GPIO5 justification

GPIO5 is a strapping pins (SDIO slave timing), but that strap is irrelevant when booting from SPI flash, and it has an internal pull-up so it idles HIGH = relay OFF

Its only quirk is a possible brief output blip from the ROM bootloader before setup() runs.

On the internal Fan, a sub-second twitch at power-on is harmless – which is precisely why a fan not the grow light is assigned here.

Servo at Boot

The firmware sets the MG996R to a known neutral angle (90°) at startup.

The 5V rail must tolerate the servo's transient current, the bulk capacitor at the servo is mandatory, not optional!

Mains isolation

The grow-light relay's COM/NO contacts carry 220V and are isolated dry contacts.

That side never touches logic ground, keep its wiring physically separated and strain relieved

5V Budget

Verify worst case 5V current before running all actuators together, servo stall is the dominant transient.

Sensor corrosion

The FC-28 and water-level probes corrode under continuous DC and should not be left energised for long periods.

Documentations & References

- **Controllers**

— **ESP32-WROOM-32E** (main controller) — Espressif official datasheet: the specifications of ESP32-WROOM-32E & ESP32-WROOM-32UE hardware, including overview, pin definitions, functional description, and electrical characteristics. https://documentation.espressif.com/esp32-wroom-32e_esp32-wroom-32ue_datasheet_en.html — this is the one to cite for your strapping-pin and ADC facts; note the datasheet confirms ESP32 has five strapping pins: MTDI, GPIO0, GPIO2, MTDO, GPIO5, which backs your GPIO 5 exception. mitEspressif

— **ESP32-CAM** (AI-Thinker) — official product page: https://docs.ai-thinker.com/en/esp32-cam · datasheet PDF: https://www.universal-solder.ca/downloads/ESP32-CAM.pdf · its camera, the **OV2640**, has a separate datasheet

- **Sensors**

— **BMP280** — Bosch Sensortec official datasheet (the primary source): https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmp280-ds001.pdf. Confirms your I²C facts — supports I2C (up to 3.4 MHz) and SPI (up to 10 MHz) digital interfaces and 1.71V~3.6V supply. LCSC ElectronicsBosch Sensortec

— **DHT11** — Aosong (manufacturer) manual, hosted via Mouser: https://www.mouser.com/datasheet/2/758/DHT11-Technical-Data-Sheet-Translated-Version-1143054.pdf · cleaner reference page: https://components101.com/sensors/dht11-temperature-sensor

— **MQ135** — Winsen (manufacturer) official manual: https://www.winsen-sensor.com/d/files/PDF/Semiconductor%20Gas%20Sensor/MQ135%20(Ver1.4)%20-%20Manual.pdf. This documents the **heater/load-resistor circuit** behind your 5 V + divider decision — the sensor requires two voltage inputs: heater voltage (VH) and circuit voltage (VC). Texas Instruments

— **Soil Moisture (FC-28 + LM393)** — module reference: https://components101.com/modules/soil-moisture-sensor-module (for the LM393 comparator, TI's LM393 datasheet is the primary source).

— **LDR module (KY-018)** and **water-level sensor** — these are commodity modules with no manufacturer datasheet; cite a module reference page (components101 / lastminuteengineers) rather than a datasheet, and note in your doc that the *specs are module-level, not chip-level*.

- **Actuators**

— **MG996R servo** — TowerPro (manufacturer) page: https://towerpro.com.tw/product/mg996r/ · datasheet PDF: https://www.handsontec.com/dataspecs/motor_fan/MG996R.pdf. Confirms your numbers — operating voltage range of 4.8-7.2V and Stall torque : 9.4kg/cm (4.8v); 11kg/cm (6.0v). RadioLocmanDatasheetsPDF.com

— **8-Channel Relay — SONGLE SRD-05VDC-SL-C** — datasheet PDF: https://www.handsontec.com/dataspecs/relay/SRD-05VDC-SL-C.pdf. This backs your coil-current budget line precisely: a 5 VDC coil with a rated power consumption of 0.36 W, corresponding to a nominal coil current of approximately 71.4 mA — i.e. your "~80 mA per coil" estimate is spot-on, and the pull-in voltage is 75% maximum of the rated voltage is exactly why you feed JD-VCC a full 5 V. RadioLocmanRadioLocman

— **Fans / pump / humidifier / LED grow light** — generic loads; no canonical datasheets. Cite the vendor listing you bought them from for current ratings (which matters for your §5.3 budget).

- **Power & Interface Ics**

— **LM2596** (buck regulator) — Texas Instruments official: https://www.ti.com/product/LM2596. Confirms SIMPLE SWITCHER® 4.5V to 40V, 3A Low Component Count Step-Down Regulator — your 5 V→3.3 V use is well inside range. Olimex

— **PCF8574** (the LCD's I²C backpack driver) — TI datasheet: REMOTE 8-BIT I/O EXPANDER FOR I2C BUS, https://www.ti.com/lit/ds/symlink/pcf8574.pdf. Cite this for your 0x27/0x3F addressing (the PCF8574 vs PCF8574A base-address split). ESP Boards

— **S-25-5 PSU**, **HW-411/LM2596 module board**, **bidirectional level converter (BSS138)** — for the level converter, cite onsemi's **BSS138** datasheet; the S-25-5 and HW-411 are generic boards (cite vendor spec sheets).